

Creative Ownership in the Age of AI

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EC 2026

Introduction

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- Novels

The AI-generated Books Trend is Getting Worse

It has gotten so bad that Amazon's new policy limits self-published Kindle ebooks to three a day.



Sascha Brodsky, Contributor
January 24, 2024

⌚ 4 Min Read

AI BUSINESS

Introduction

Generative AI systems—trained on large corpora of human-authored works—are increasingly used to produce new creative content.

- Novels
- Cartoons

Generating New Yorker-Style Cartoons with AI

Harnessing the power of OpenAI's ChatGPT to Transform Cartoon ideas into Reality



Georgiy Ermakov

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5 min read · Oct 25, 2023

Introduction

Generative AI systems—trained on large corpora of human-authored works—are increasingly used to produce new creative content.

- Novels
- Cartoons
- Even AI “actors”

Tilly Norwood: how scared should we be of the viral AI ‘actor’?

A bunch of code is being pushed as the next Scarlett Johansson, a creation that is already causing pushback from real human actors

Tilly Norwood



A 2025 AI-generated image of Tilly Norwood, taken from the character's [Instagram](#) account

Introduction

Generative AI systems—trained on large corpora of human-authored works—are increasingly used to produce new creative content.

- Novels
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These creations can closely imitate style without copying content, a capability now central to ongoing litigation.

- Andersen v. Stability AI
 - Disney Enterprises v. Midjourney
 - Universal Music Group Recordings v. Suno
- + many others

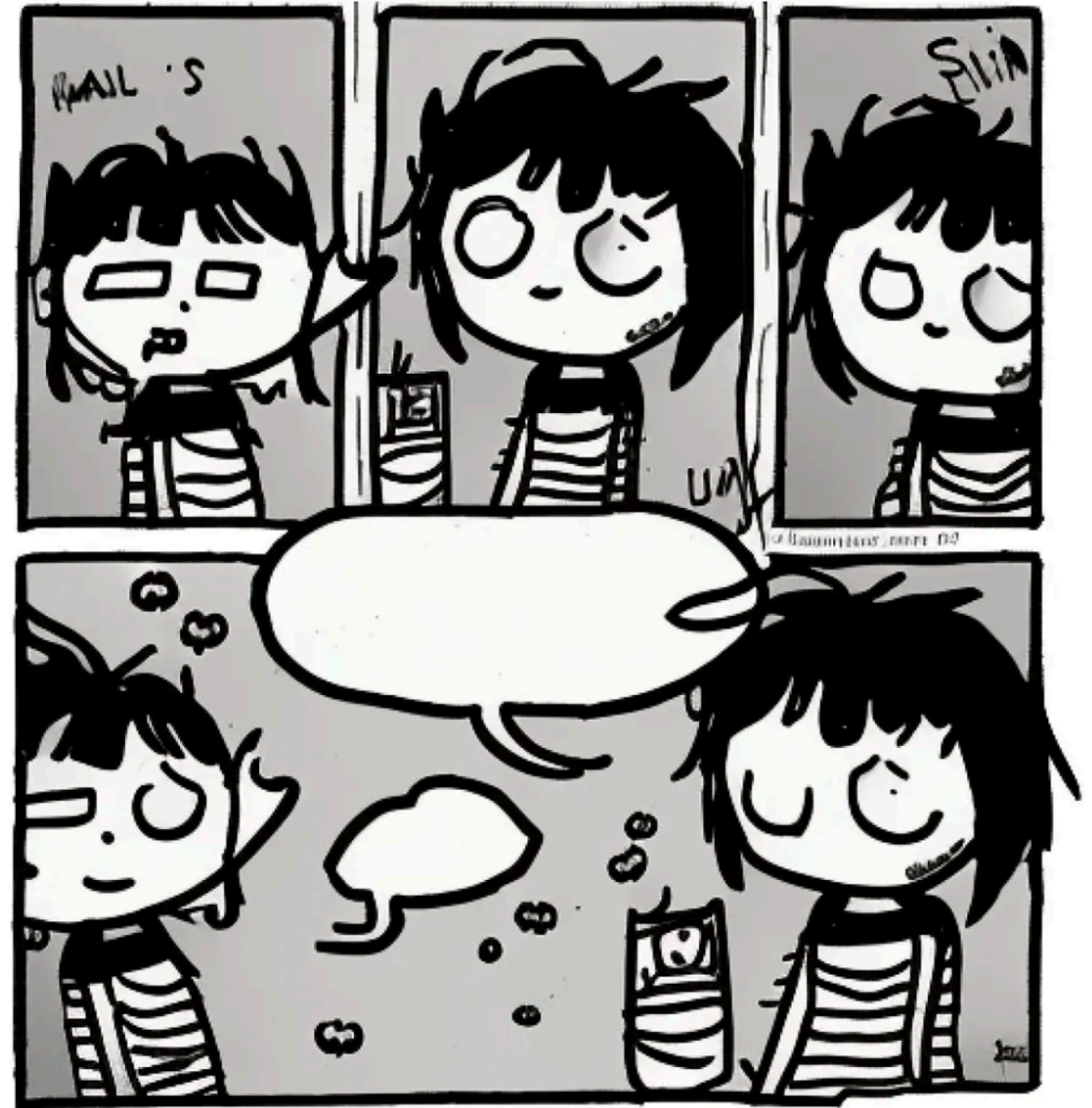
Introduction

A recent example of my work



Sarah Andersen, *NYTimes*, 2022

An A.I. generated image that I created when I used my name as a prompt



It's not perfect — but it has captured the signature elements of my drawing style. The black bangs, striped shirt and wide eyes are immediately recognizable. As A.I. gets increasingly sophisticated, I am growing more concerned about what's possible.

Legal Context

Existing copyright doctrine restricts both the inputs and outputs of GenAI:

1. **Training** on copyrighted works may violate copyright law.

- Possibly too strong: human authors learn by reading copyrighted books.
- May overly constrain AI capabilities to not allow it to learn in a similar way
- Moreover, fair use doctrine suggests that training on copyrighted works may be OK if the use is sufficiently “transformative”

2. **Generated outputs** may violate copyright law if they are “substantially similar” to an existing work.

- Copyright law has historically drawn a line between protectable **expression** and unprotectable **style** to preserve creative competition and follow-on innovation.
- But did not account for the emergence of AI systems that could easily, flexibly, quickly mimic style

This Paper

Existing notions of copyright and intellectual property **may not be appropriate for a post-generative AI world.**

We propose a new criterion for evaluating when an output is derivative of an existing work:

Would the output have been generable without that existing work in its training corpus?

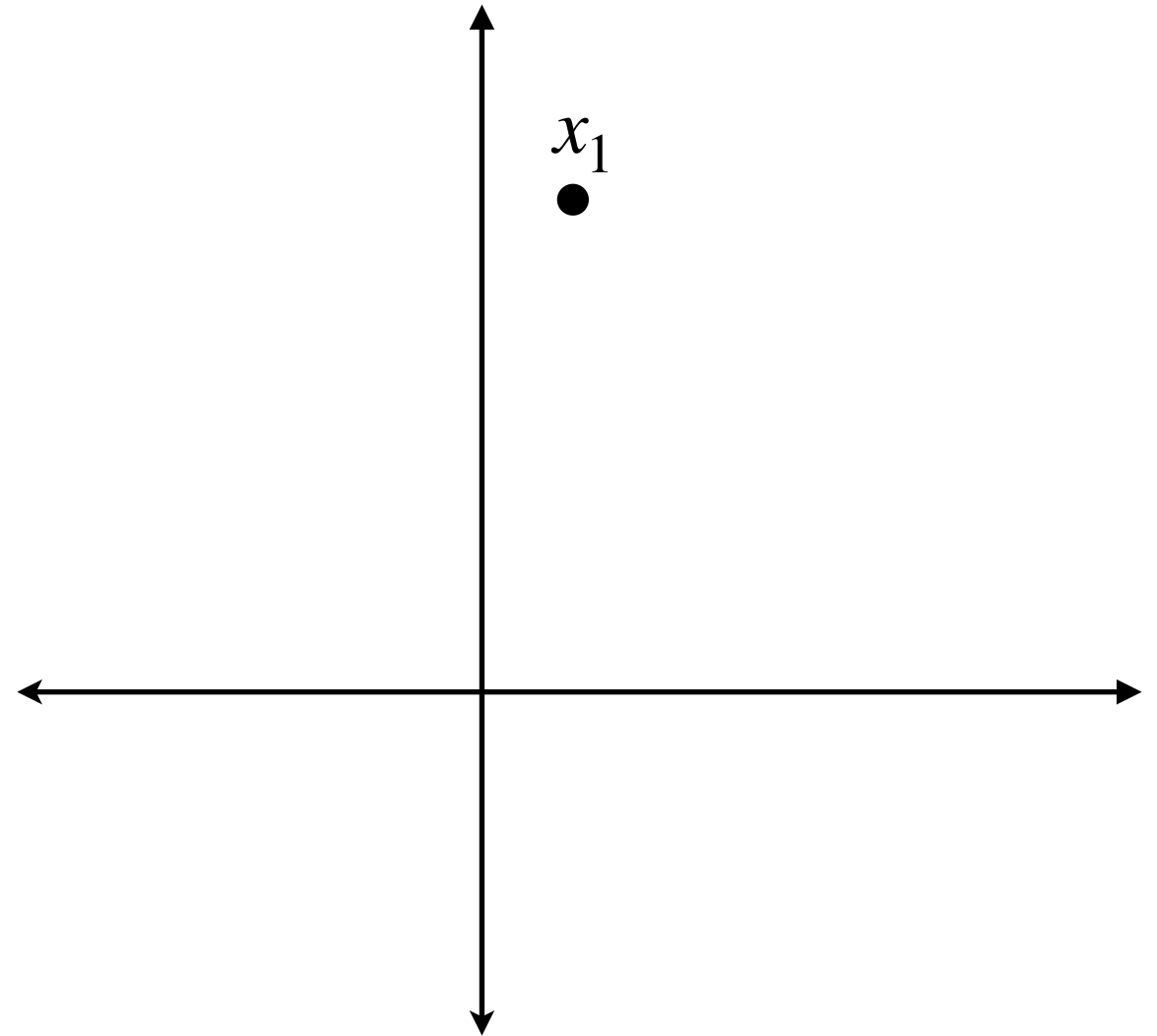
Study the implications of this criterion for permissible generation

- Whether the creative process is **light or heavy tailed** matters critically for the restrictiveness of this criterion

Model

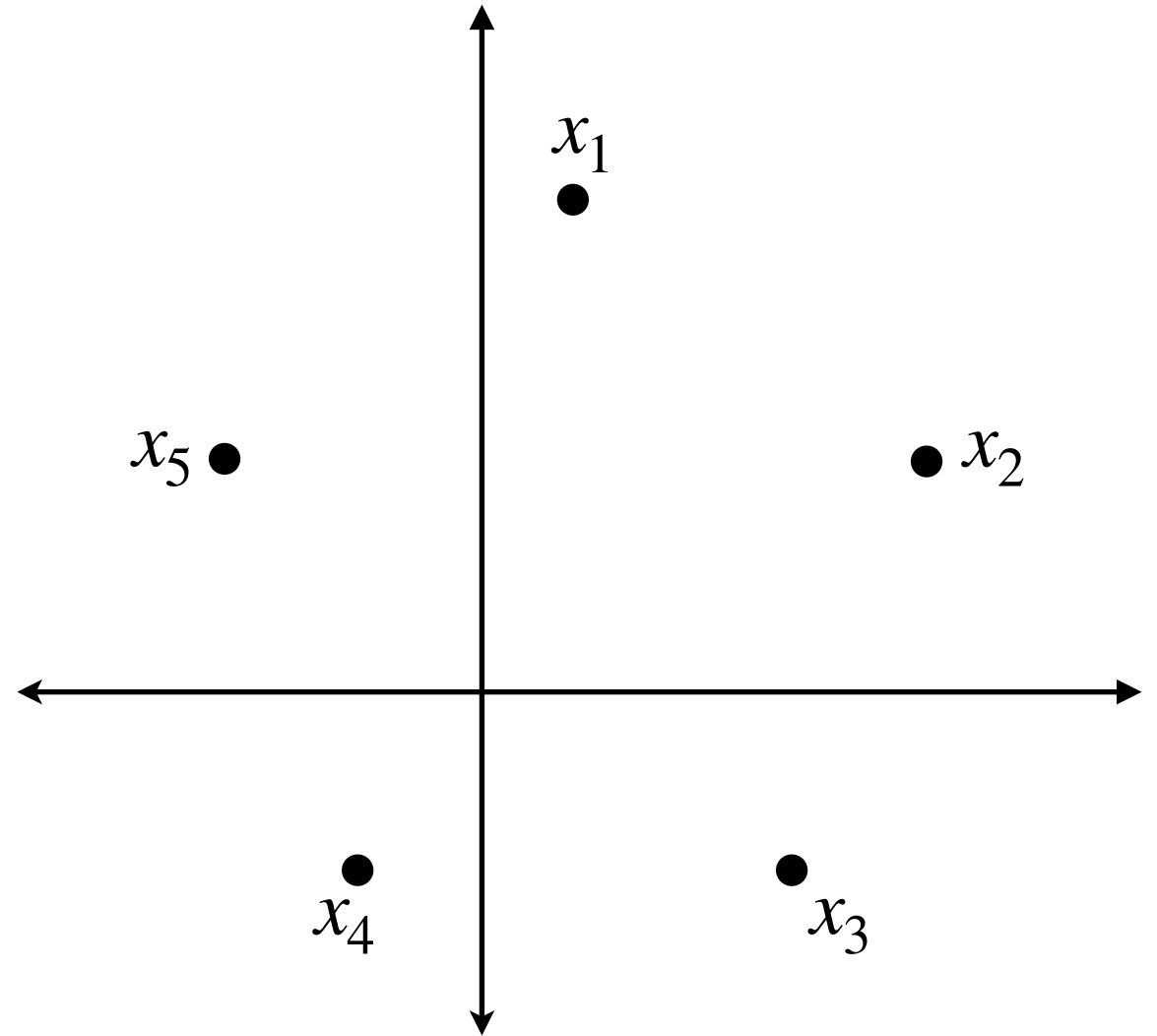
Model

- A *creation* is a point $x \in \mathbb{R}^d$



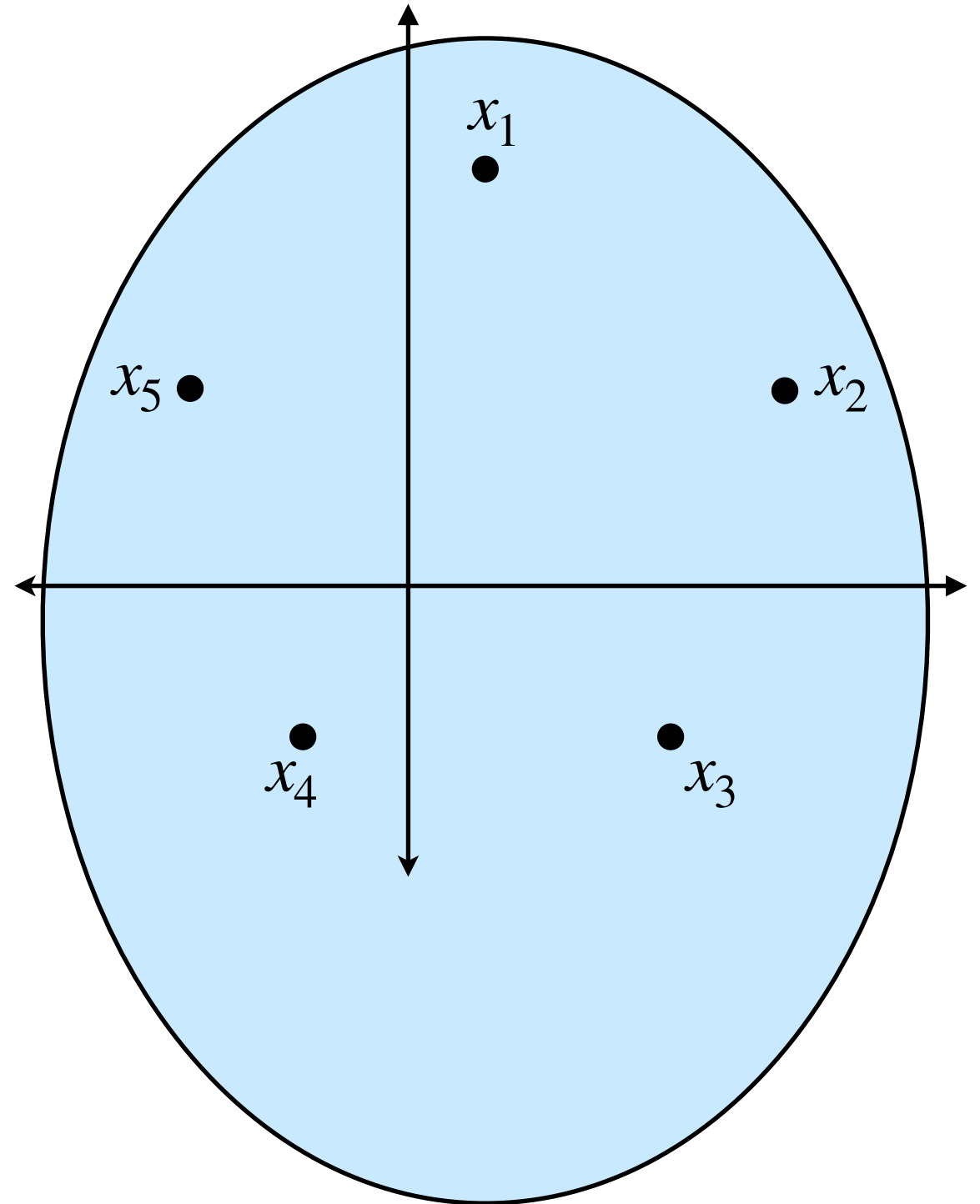
Model

- A *creation* is a point $x \in \mathbb{R}^d$
- A *corpus* is a set of creations



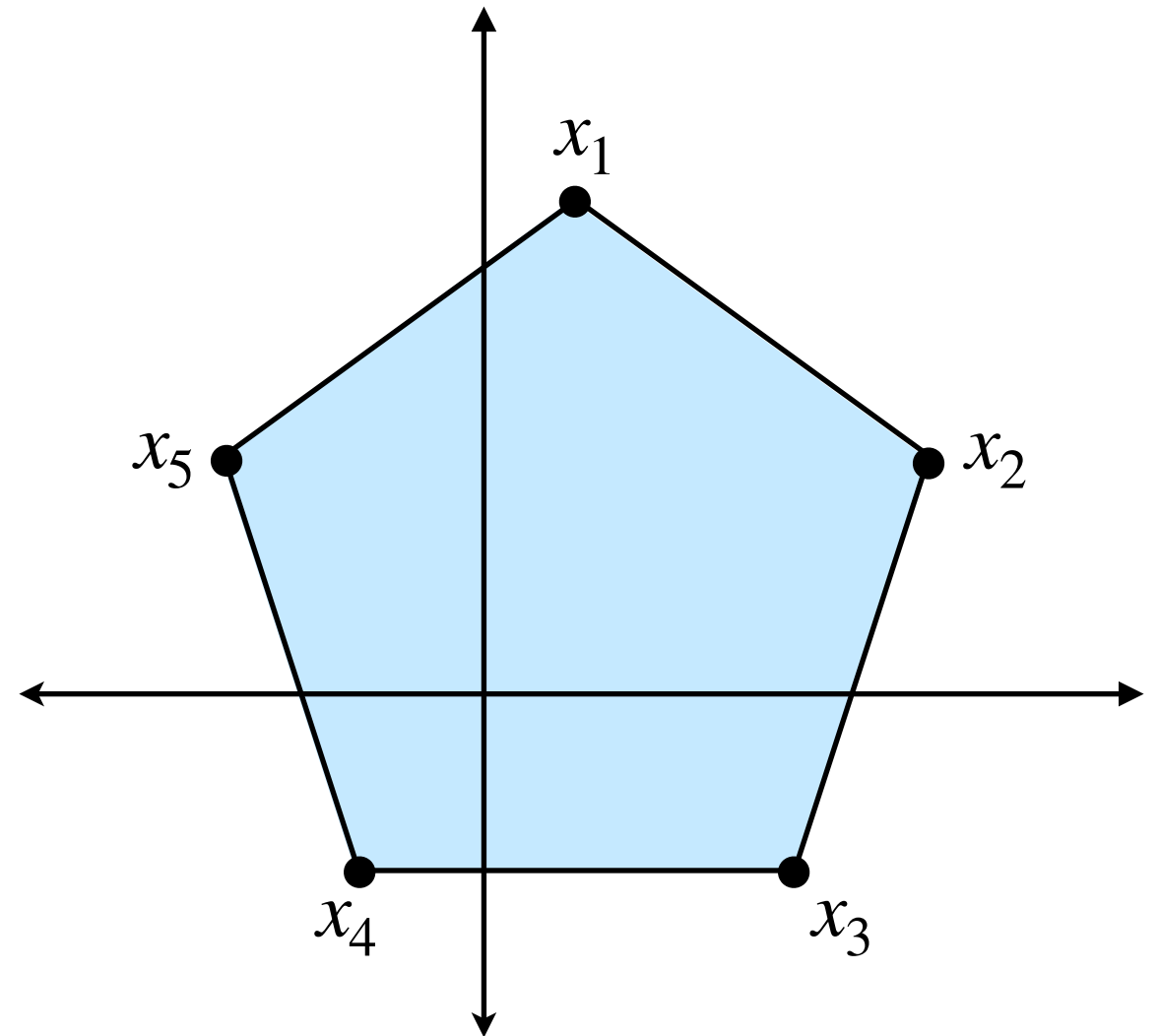
Model

- A *creation* is a point $x \in \mathbb{R}^d$
- A *corpus* is a set of creations
- A *generator* is a technology for producing new creations from an existing corpus, satisfying:
 1. $C \subseteq g(C)$
 2. $C \subseteq C'$ implies $g(C) \subseteq g(C')$
 3. $g(g(C)) = g(C)$



Model

- A *creation* is a point $x \in \mathbb{R}^d$
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 1. $C \subseteq g(C)$
 2. $C \subseteq C'$ implies $g(C) \subseteq g(C')$
 3. $g(g(C)) = g(C)$
- The generator is *convex-valued* if every $g(C)$ is convex (minimal case is the convex-hull generator)



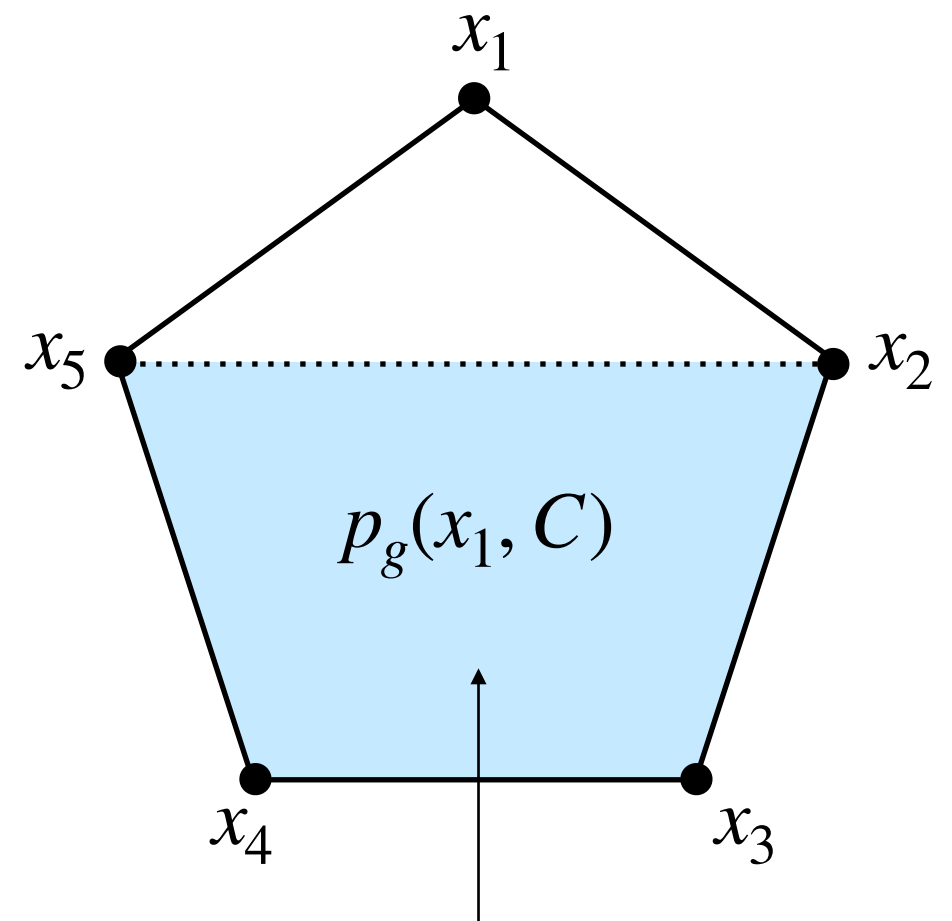
Permissible Generation

Definition: Fix any generator g , corpus C , and work $x \in C$. The x -**permissible set** is

$$p_g(x, C) := g(C \setminus \{x\})$$

i.e., all creations that could have been generated without x

Convex hull generator:



x_1 -permissible set

(generable creations that do not require x_1)

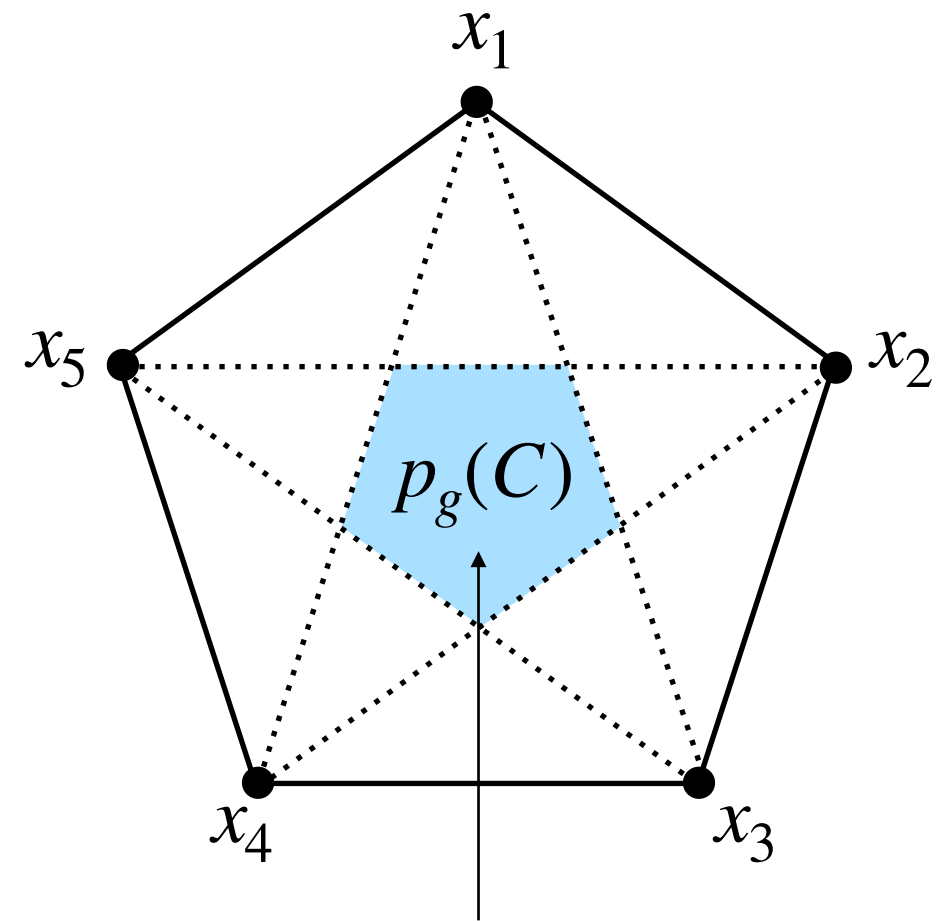
Permissible Generation

Definition: Fix any generator g and corpus C . The **permissible set** is

$$p_g(C) := \bigcap_{x \in C} p_g(x, C)$$

i.e., all creations that do not critically depend on any $x \in C$

Convex hull generator:



permissible set

(generable creations that do not critically rely on any $x \in C$)

Discussion

Q: What does it mean for an existing work to “have a claim”?

- The predecessor may be entitled to **attribution**.
- The predecessor may be entitled to **financial compensation or rent**.
- The downstream work may be **restricted or prohibited**.

Our framework is agnostic about which.

Discussion

Q: What does it mean for an existing work to “have a claim”?

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Q: How would permissibility be evaluated in practice?

- Brute-force retraining of an algorithm is not feasible
- Machine unlearning provides methods to approximate the role of specific training points (Bourtoule et al., 2019; Xiong et al., 2024)
- Can also view permissibility as an evidentiary question and compare outputs across independently trained algorithms

The Structure of the Permissible Set

- **Result:** The permissible set is nonempty under a weak condition (the size of the corpus exceeds the *Radon number of the generator*)
 - When the generator is convex-valued, $d + 2$ points is sufficient
- **Result:** The permissible set is self-stable: $g(p_g(C)) = p_g(C)$, i.e., combining permissible works cannot yield a violation
- **Result:** Adding works to the corpus expands the permissible set (i.e., $p_g(C)$ is monotone)

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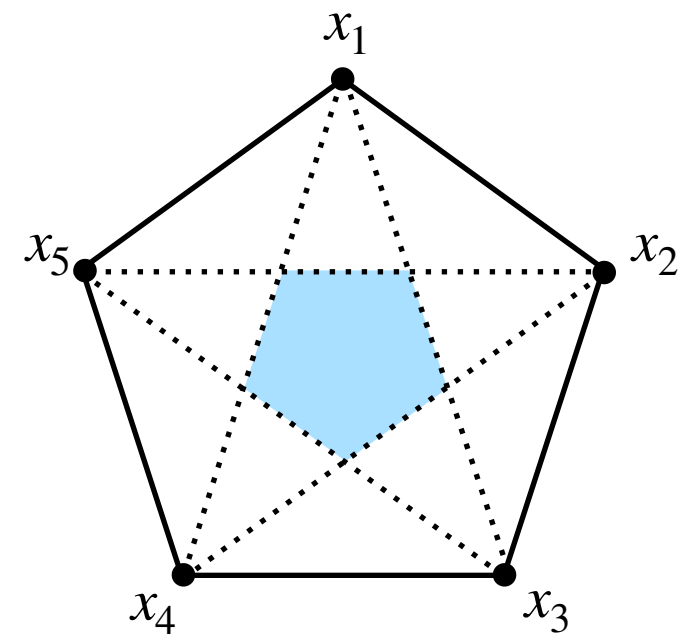
We next consider what the permissible set looks like when the corpus is large—will individual creations retain their importance?

Permissible Ratio

Let $Vol(A)$ denote the Lebesgue measure (or *volume*) of the set $A \in \mathcal{C}$.

Definition: The **permissible ratio** is:

$$r_g(C) := \frac{Vol(p_g(C))}{Vol(g(C))}$$

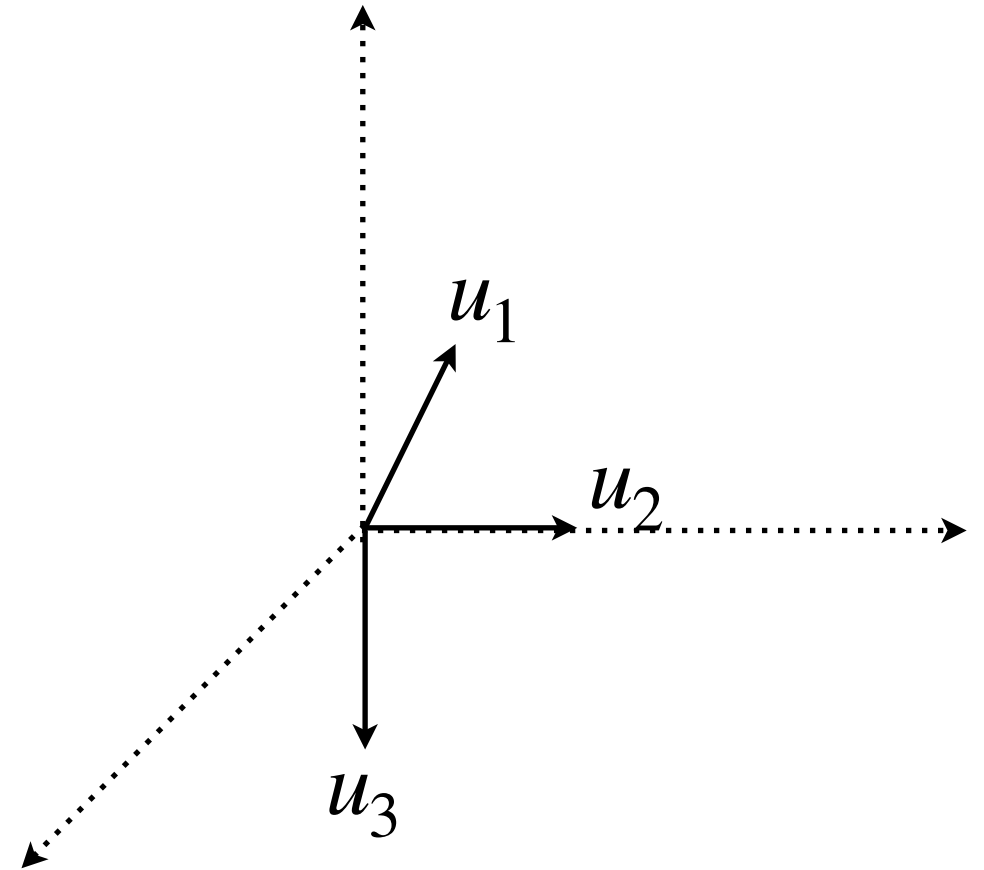


i.e., the proportion (by volume) of generable creations that are permissible.

Corpus Formation

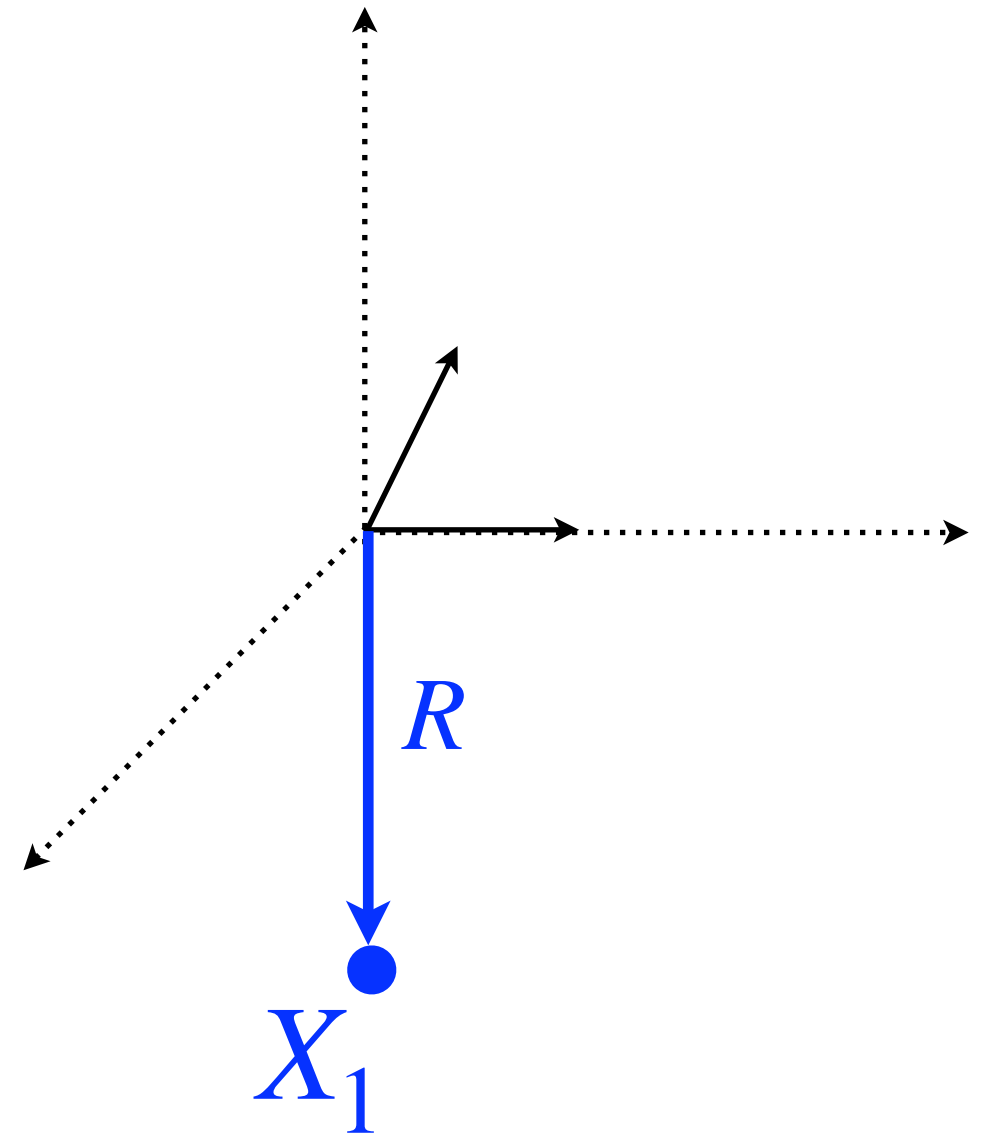
Exactly d directions is a simplifying assumption for the talk, can generalize with additional conditions

- Fixed creative directions $u_1, \dots, u_d$ (unit vectors) spanning the creative space $\mathbb{R}^d$



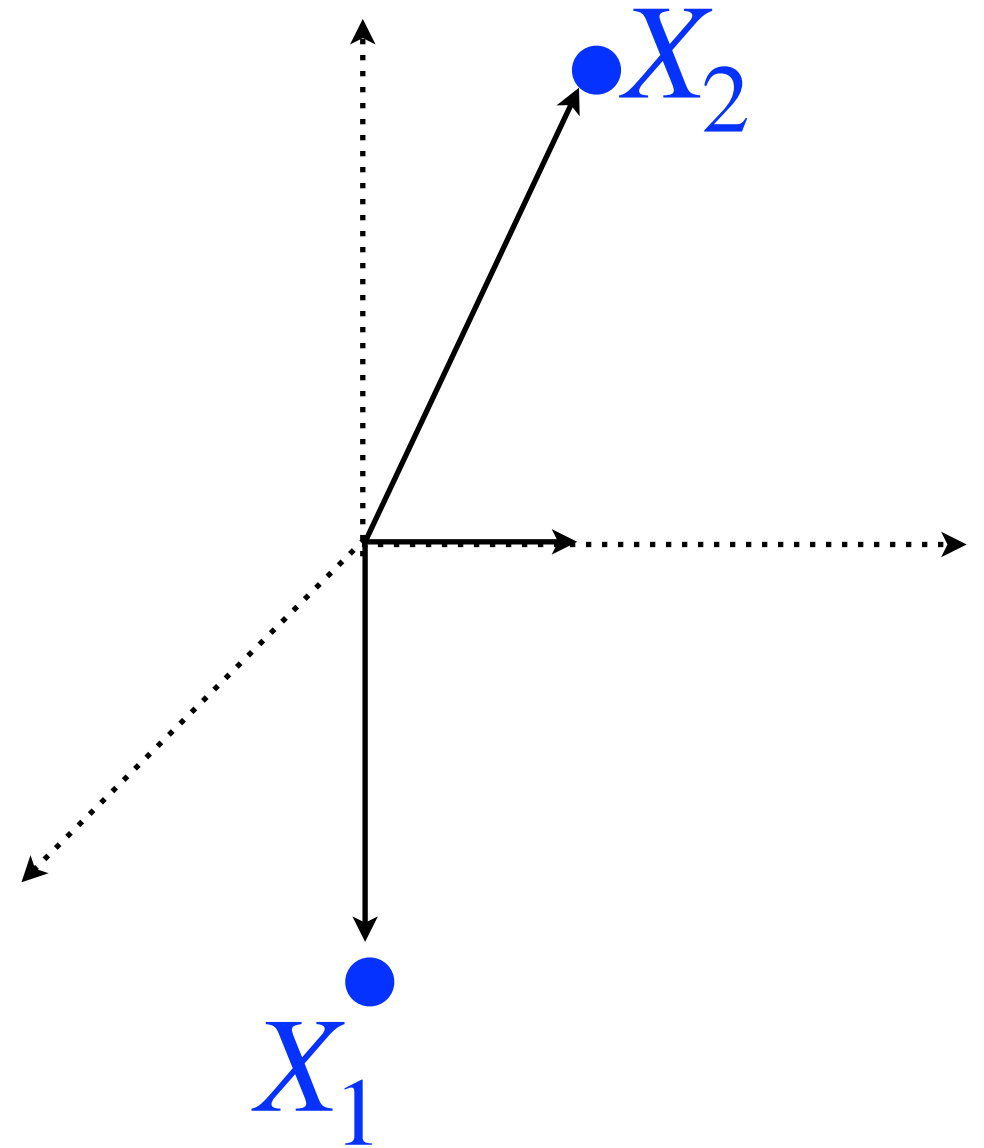
Corpus Formation

- Fixed creative directions $u_1, \dots, u_d$ (unit vectors) spanning the creative space $\mathbb{R}^d$
- At each date n :
 - Choose a direction $k \in \{1, \dots, d\}$ (possibly given the history)
 - Draw a magnitude $R \in \mathbb{R}_+$ from the distribution F_k
- The new creation is $X_n = Ru_k$



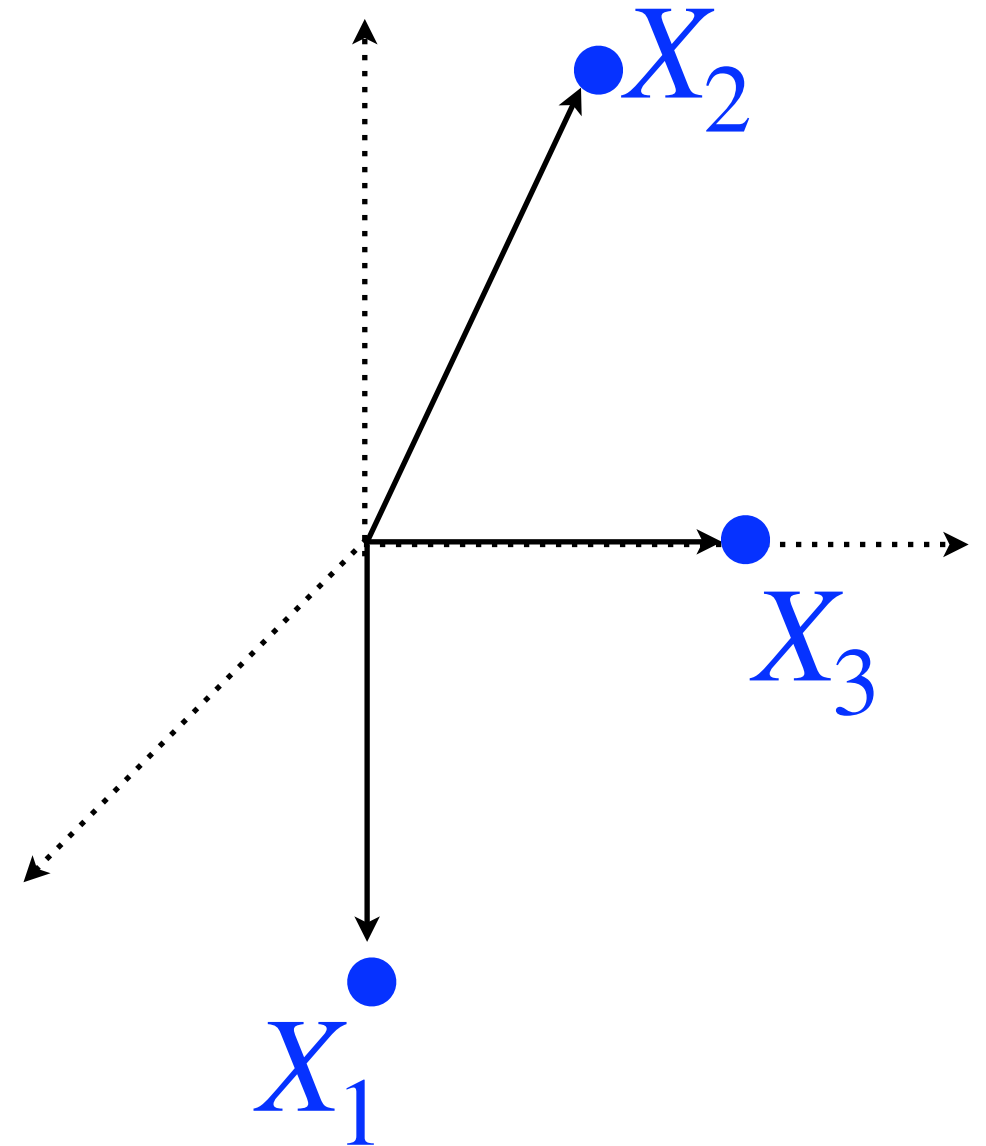
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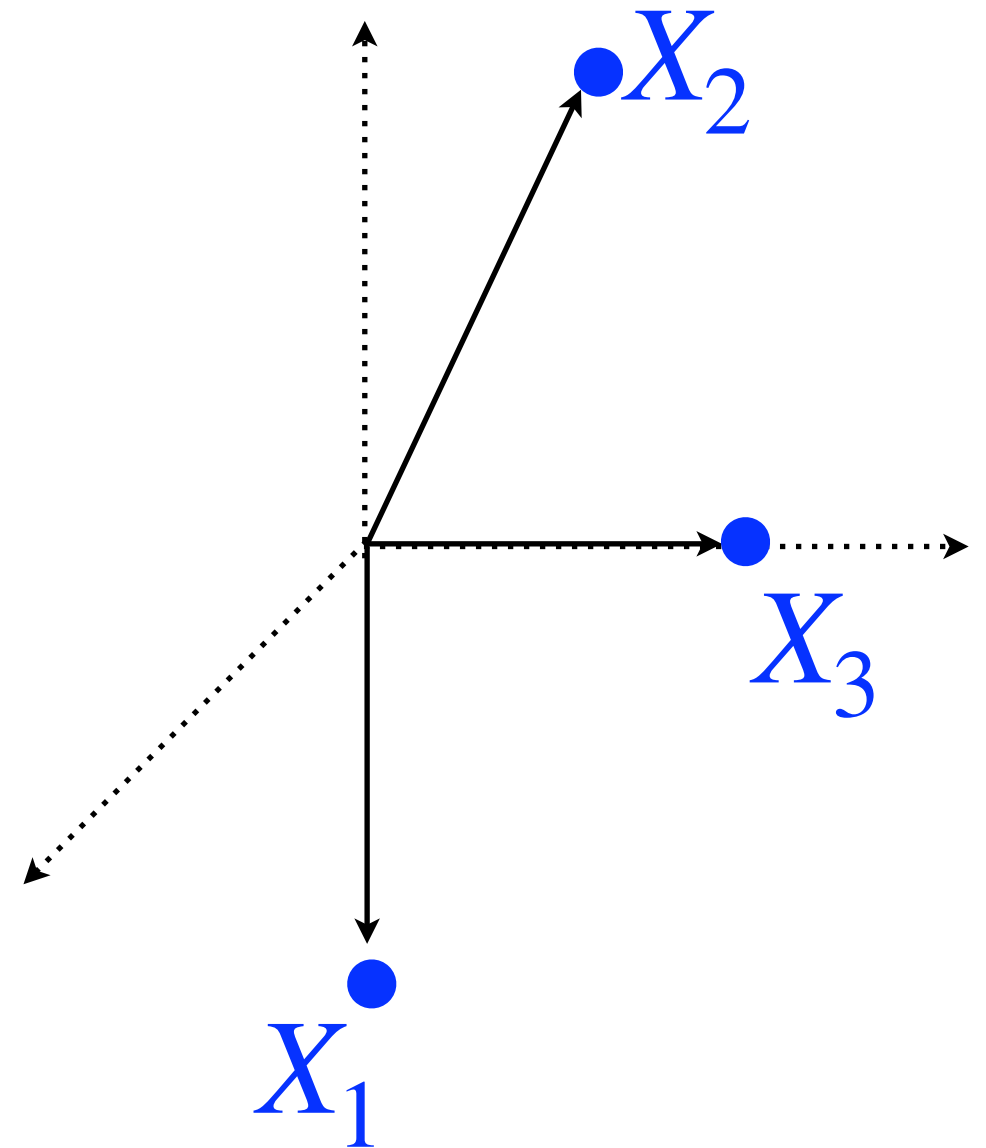
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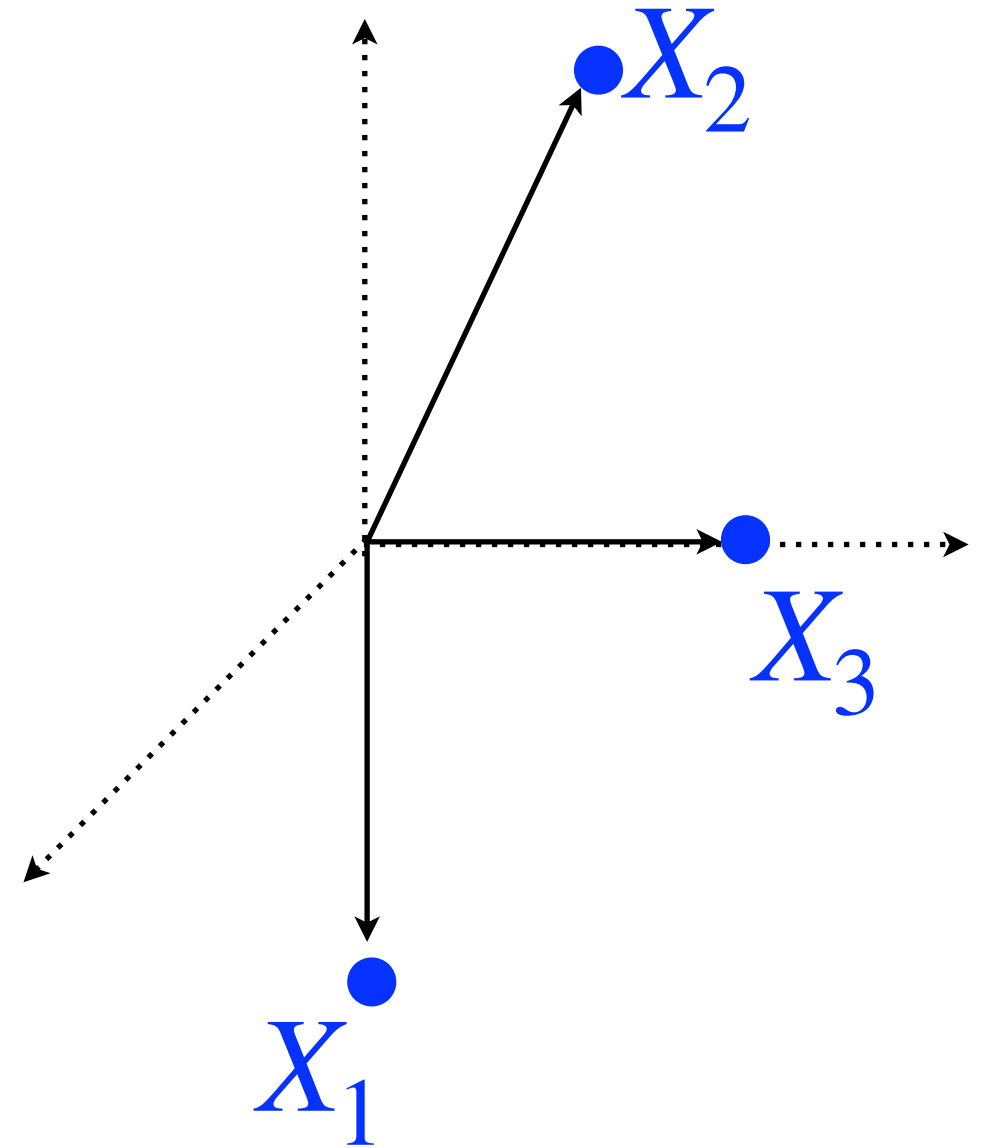
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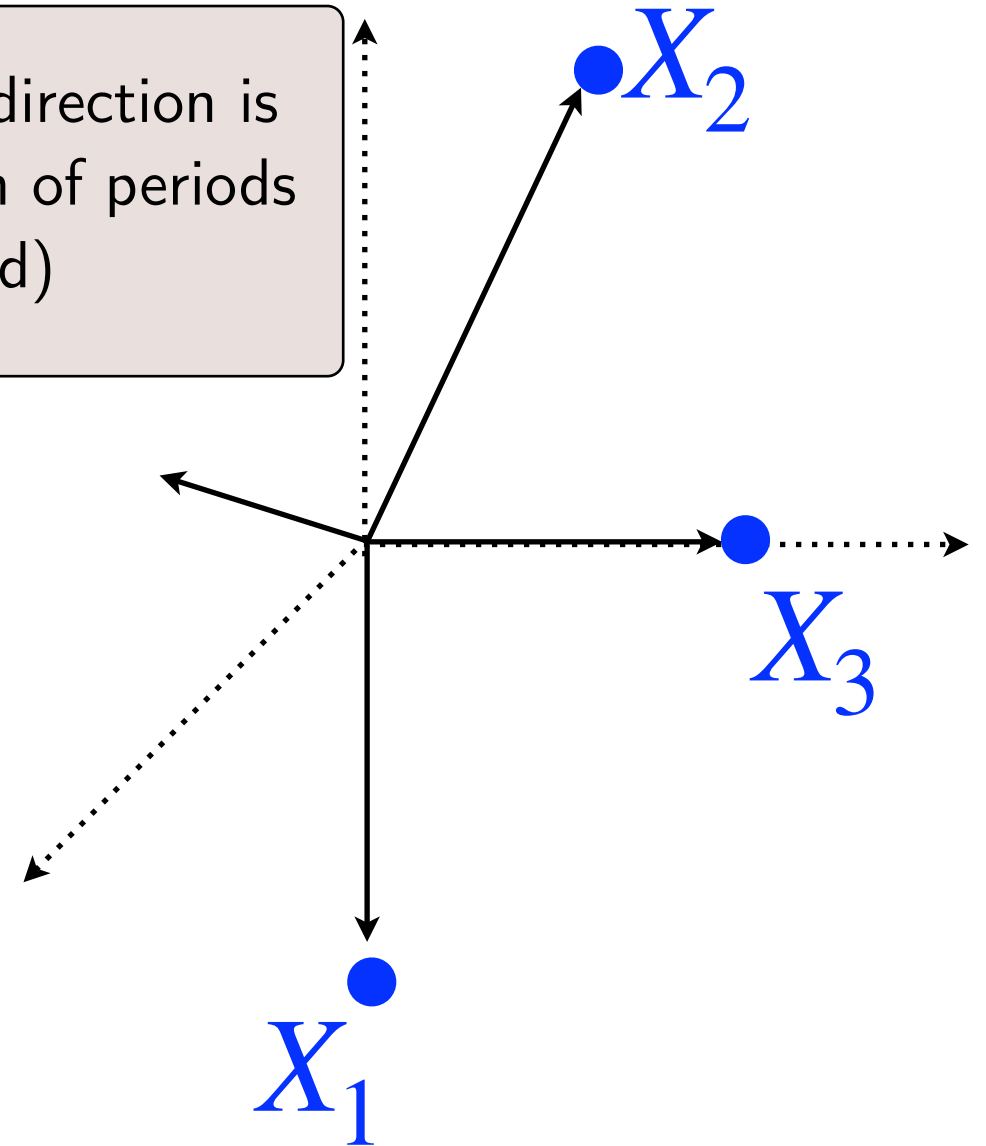


Question: What is $\lim_{n \rightarrow \infty} r_g(C_n)$, i.e. the fraction of generable works are eventually permissible?

Corpus Formation

- Fixed creative directions (vectors) spanning the space
- At each date n :
 - Choose a direction $k \in \{1, \dots, d\}$ (possibly given the history)
 - Draw a magnitude $R \in \mathbb{R}_+$ from the distribution F_k
- The new creation is $X_n = Ru_k$
- The date- n corpus is $C_n = \{X_1, \dots, X_n\}$

For the talk, suppose each direction is chosen in a positive fraction of periods (stronger than what we need)



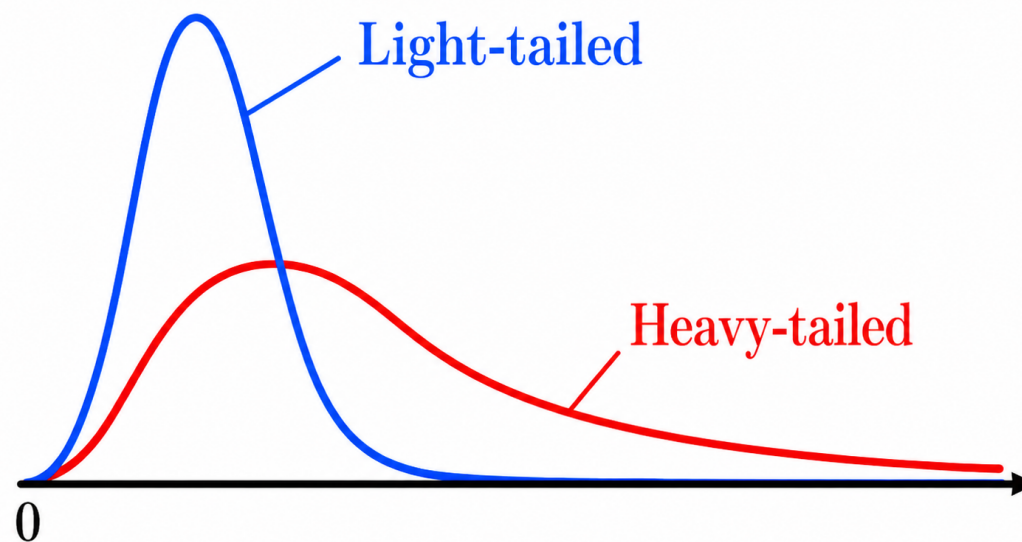
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Two Tail Regimes

Definition: Say that a distribution F on $\mathbb{R}_+$ has

- **light tails** if $\bar{F}(ct) \leq \bar{F}(t)^c$ for all $c \geq 1$ and $t \geq 0$ (e.g., truncated Gaussian)
- **heavy tails** if there is a constant α such that $\bar{F}(ct) \geq c^{-\alpha} \bar{F}(t)$ for all $c \geq 1$ and $t \geq 0$ (e.g., Pareto)

where $\bar{F} = 1 - F$ denotes the survival function.



Main Result

Theorem (informal): Fix a convex-valued, continuous, and bounded generator g .

A. If each F_k has light tails, then

$$\lim_{n \rightarrow \infty} r_g(C_n) = 1$$

B. If at least one F_k is heavy-tailed, then:

$$\liminf_{n \rightarrow \infty} r_g(C_n) = 0$$

Takeaways

- In creative markets where innovation is light-tailed, individual creators are eventually marginal
- If there are rare but significant breakthroughs,
 - individual creators remain pivotal
 - our criterion strongly binds even with arbitrarily large corpora

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- In creative markets where innovation is light-tailed, individual creators are eventually marginal
- If there are rare but significant breakthroughs,
 - individual creators remain pivotal
 - our criterion strongly binds even with arbitrarily large corpora
- We've taken the regime as given, but could also consider the question of which we would expect to emerge endogenously
- e.g., incentives for creators might push towards a heavy-tailed regime (especially if creative protections follow our proposal)

Generalization to Protected Groups

- Copyright disputes sometimes arise at the level of collections rather than single works



Generalization to Protected Groups

- Copyright disputes sometimes arise at the level of collections rather than single works
- Can generalize our criterion to handle this
 - New primitive: a family of sets of works $\{A_1, \dots, A_K\}$ where each A_i is a **protected group**
 - Define the permissible set to be the intersection of the A_i -permissible sets (rather than the x_i -permissible sets)
- Our main result here: violations are **superadditive**
- If a firm that owns A_1 merges with a firm that owns A_2 , the permissible set of generations becomes smaller

Conclusion and Open Questions

- Our paper proposes a new definition for when a generated work is “derivative” of an original
- Directions for extension:
 - Endogenizing the generator and the distribution of creative works.
 - Suppose creators are paid as a function of how much their works expand the generable set. What works would be created? What coalitions optimally form?
 - Cardinal measures of influence

Thank You